Reg. No. | | | | | | | | | | | [17BS4CS02]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

STATISTICS AND PROBABILITY-II

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions, choosing at least **TWO** questions from each part.
2. Any missing data may be suitably assumed.
3. Statistical tables will be provided.

PART - A

1. a. The probability distribution of a finite random variable X is given by the following table. **08**

X	-2	-1	0	1	2	3
P(X)	0.1	K	0.2	2K	0.3	K

Find the value of K, mean and variance.

b. The marks of 1000 students in an examination follows a normal distribution with mean 70 and standard deviation 5. Find the number of student whose marks will be ; **06**

i. Less than 65

ii. More than 75 [Given $\Phi(1) = 0.3413$].

2. a. Define; **08**

i. Type - I & Type –II errors

ii. Null Hypothesis & Alternate Hypotheis

iii. Random sampling with replacement & without replacement

iv. One tailed and two tailed tests

b. Distinguish between the 'Census' and 'Sampling' methods of collection of data and compare their merits and demerits. **06**

3. a. Certain tubes manufactured by a company have mean life time of 800 hours and standard deviation of 60 hours. Find the probability that a random sample of 16 tubes taken from group will have a mean life time **08**

i. Between 790 hours and 810 hours

ii. Less than 785 hours. [Given $\Phi(0.67) = 0.2486$ & $\Phi(1) = 0.3413$]

b. A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured third class, 90 had secured second class and 20 had secured first class. Do these figures support the general examination result which is in the ratio 4 : 3 : 2 : 1 for the respective categories. **06**

[Use $\chi^2_{0.05} = 7.81$ for 3 d.f.]

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4. a. Fit a second degree parabola $y = a + bx + cx^2$ which fits most closely with the observations **08**

x	-3	-2	-1	0	1	2	3
y	4.63	2.11	0.67	0.09	0.63	2.15	4.58

b Psychological tests of intelligence and of engineering ability were applied to 10 students. The following data shows the intelligent ratio and engineering ratio. Calculate the coefficients of correlation **06**

Student	A	B	C	D	E	F	G	H	I	J
IR	105	104	102	101	100	99	98	96	93	92
ER	101	103	100	98	95	96	104	92	97	94

PART - B

5. a. Find the multiple linear regression equation of X_1 on X_2 and X_3 from the data relating to three variables given below: **08**

X_1	4	6	7	9	13	15
X_2	15	12	8	6	4	3
X_3	30	24	20	14	10	4

b. The table shows the corresponding values of three variables, X_1, X_2 and X_3. Find the least square regression equation of X_3 on X_1 and X_2 **06**

X_1	3	5	6	8	12	14
X_2	16	10	7	4	3	2
X_3	90	72	54	42	30	12

6. a. Ten samples each of size 5 are drawn at regular intervals from a manufacturing process. The sample means ($\bar{X}$) and their ranges (R) are given below: **08**

Sample Number	1	2	3	4	5	6	7	8	9	10
Mean ($\bar{X}$)	49	45	48	53	39	47	46	39	51	45
Range (R)	7	5	7	9	5	8	8	6	7	6

Calculate the control limits in respects of $\overline{X}$ - chart and R – Chart (Given: $A_2 = 0.58$, $D_3 = 0$, $D_4 = 2.15$). Comment on the state of control.

b. A tyre manufacturer randomly selected 40 tyres at the end of each shift to test for tyres that are defective. The number of defectives in 12 shifts is as follows. 4, 2, 0, 5 , 2 , 3 , 14, 2 , 3, 4, 12 , 3. Construct a P- chart for this process. Is the production process under control?

7. a. Show that $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ is a regular stochastic matrix. Also find the associated unique fixed probability vector. **08**

b. A student's study habits are as follows. If he studies one night, he is 70% sure not to study the next night. On the other hand, if he does not study one night, he is 60% sure, not to study the next night. In the long run how often does he study. **06**

8. a. Define Simulation and system simulation by giving suitable example. Write the components of the system. **08**

b. The statistics of timings taken over a month of arrival of a train in a particular station is as follows **06**

Time (T)	9AM	9.05	9.10	9.15
Days (P)	6	12	7	5

Simulate this problem for 7 days starting from Monday.